Comprehensive Physical Analysis and Constraint Mechanics of Anharmonic Quantum Otto Efficiency Plots

Detailed Explanation of $\eta(\eta_C)$ Behavior and Comparison with Literature (`optimal-performance.pdf`)

Anharmonic Quantum Otto Engine Study

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Contents

1 Introduction

In quantum thermodynamics, the thermal efficiency $\eta = W_{\text{ext}}/Q_h$ of a quantum heat engine operating between a cold bath at inverse temperature $\beta_c = 1/(k_B T_c)$ and a hot bath at inverse temperature $\beta_h = 1/(k_B T_h)$ is fundamentally bounded by the Carnot efficiency limit:

$$\eta_C = 1 - \frac{\beta_h}{\beta_c} = 1 - \frac{T_c}{T_h} \tag{1}$$

When analyzing the performance of our anharmonic spin-1 NV-center quantum Otto engine, two distinct plotting regimes emerge depending on how operating parameters are constrained:

1. **Fixed-Frequency Baseline Regime:** Operating frequencies (ω_c, ω_h) are held fixed at baseline values (e.g. $R_\omega = \omega_h/\omega_c = 7.0$). In this regime, engine operation ($W_{\text{ext}} > 0$) occurs exclusively in the high-gradient region $\eta_C \in [0.84, 0.96]$.

2. **Variable / Re-optimized Frequency Regime:** Frequencies (ω_c, ω_h) are dynamically re-optimized for each individual η_C. In this regime, the efficiency curves start at $\eta_C = 0$ and extend across the full domain $[0, 0.96]$, matching standard unconstrained literature models such as Fig. 2 of `optimal-performance.pdf`.

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2 Detailed Physics of Plot Set 1 and Plot Set 2

2.1 Plot Set 1: 2x2 Grid Grouped by Case

In Plot Set 1 (`eta_vs_carnot_grid_by_case.pdf`), each subplot corresponds to one specific stroke transition model (Cases 1 to 4), and contains 5 distinct curves representing different anharmonicity parameters $\alpha \in [0.00, 0.20]$:

- **Case 1 (Symmetric, $\lambda_{ab} = \lambda_{cd} = 1$):** Quantum transitions are purely adiabatic without non-adiabatic excitation losses. The efficiency curve approaches the ideal ratio $\eta = 1 - \omega_c/\omega_h = 1 - 1/7.0 \approx 0.8571$.

- **Case 2 (Hot-stroke Asymmetric, $\lambda_{ab} > 1, \lambda_{cd} = 1$):** Quantum friction occurs during the hot expansion stroke. Efficiency is depressed below the ideal line as friction consumes part of the work produced.

- **Case 3 (Cold-stroke Asymmetric, $\lambda_{ab} = 1, \lambda_{cd} > 1$):** Non-adiabatic excitation occurs during the cold compression stroke. Cold-side friction directly increases heat rejection into the cold bath, degrading efficiency more severely than hot-stroke friction.

- **Case 4 (Dual Asymmetric, $\lambda_{ab} > 1, \lambda_{cd} > 1$):** Both strokes suffer from non-adiabatic excitation losses, leading to the lowest efficiency across all α.

2.2 Plot Set 2: 1x5 Layout Grouped by Anharmonicity α

In Plot Set 2 (`eta_vs_carnot_grid_by_alpha.pdf`), each panel isolates a specific value of anharmonicity α. Within each panel, the four cases are overlaid together to illustrate the hierarchy of quantum friction (Case 1 > Case 2 > Case 3 > Case 4).

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3 Why Fixed-Frequency Plots Are Restricted to $\eta_C \in [0.84, 0.96]$

3.1 1. Lower Bound Threshold ($\eta_C > 1 - 1/R_\omega$)

For a quantum engine to produce positive net work ($W_{\text{ext}} > 0$), the ratio of cold-to-hot inverse temperatures must exceed the frequency compression ratio:

$$\frac{\beta_c}{\beta_h} > \frac{\omega_h}{\omega_c} = R_\omega \tag{2}$$

Expressing β_c/β_h in terms of Carnot efficiency $\eta_C = 1 - \beta_h/\beta_c$:

$$\frac{\beta_c}{\beta_h} = \frac{1}{1-\eta_C} \tag{3}$$

Substituting into the engine condition gives:

$$\frac{1}{1-\eta_C} > R_\omega \implies 1 - \eta_C < \frac{1}{R_\omega} \implies \eta_C > 1 - \frac{1}{R_\omega} \tag{4}$$

For our NV-center system where $R_\omega = 7.0$:

$$\eta_C > 1 - \frac{1}{7.0} = \frac{6}{7} \approx 0.8571 \tag{5}$$

If $\eta_C < 0.8571$, the thermal gradient is too weak to overcome the $7.0\times$ frequency compression. The engine operates in reverse (as a heater or refrigerator), producing negative work ($W_{\text{ext}} \le 0$). Thus, no positive-work engine curves can exist below $\eta_C \approx 0.8571$ for fixed $R_\omega = 7.0$.

3.2 2. Quantum Friction Penalty ($\lambda > 1$)

In Cases 2, 3, and 4, non-adiabatic transitions generate internal quantum friction ($\lambda_{ab}, \lambda_{cd} > 1$). Overcoming this non-adiabatic excitation energy requires an even larger temperature gradient ($\eta_C \ge 0.84 - 0.89$). Below this friction threshold, $W_{\text{ext}} \le 0$, filtering out all points below $\eta_C \approx 0.84$.

3.3 3. Upper Bound Cap ($\eta_C \le 0.96$)

The upper bound is set by the diamond NV-center temperature ratio cap ($R_\beta = \beta_c/\beta_h \le 25.0$):

$$\frac{\beta_c}{\beta_h} \le 25.0 \implies \frac{1}{1-\eta_C} \le 25.0 \implies \eta_C \le 1 - \frac{1}{25.0} = 0.96 \tag{6}$$

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4 Comparison with Unconstrained Literature (`optimal-performance.pdf`)

In standard theoretical literature (such as Fig. 2 of `optimal-performance.pdf`), the compression ratio $R_\omega = \omega_h/\omega_c$ is **not fixed**. Instead, for every value of $\eta_C \in (0, 1)$, the frequencies (ω_c, ω_h) are continuously re-optimized to match the temperature gradient:

$$R_\omega(\eta_C) \approx \sqrt{\frac{\beta_c}{\beta_h}} = \frac{1}{\sqrt{1-\eta_C}} \tag{7}$$

As $\eta_C \to 0$ ($\beta_h \to \beta_c$), the optimal compression ratio scales down toward $R_\omega \to 1.0$. As a result:

- The condition $\frac{\beta_c}{\beta_h} > R_\omega$ is satisfied for **all** $\eta_C \in (0, 1)$.
- The efficiency curve traces the Carnot line η_C continuously from $\eta_C = 0$ up to 0.8571, where it hits the compression ratio cap $R_\omega \leq 7.0$ and plateaus.
- Non-ideal cases (Cases 2, 3, 4) extend continuously down to $\eta_C \approx 0.35 - 0.47$, yielding smooth curves starting near the origin, exactly as seen in Fig. 2 of `optimal-performance.pdf`.

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5 Conclusion and Recommendations for Paper Presentation

1. **Use Fixed-Frequency Active Region ($\eta_C \in [0.80, 1.00]$):** When showcasing performance under fixed operational frequencies ($R_\omega = 7.0$), set the x-axis range to $[0.80, 1.00]$ to clearly resolve the curves.
2. **Use Full-Range Re-optimized Plot ($\eta_C \in [0, 1.0]$):** When comparing against standard literature (such as `optimal-performance.pdf`), use the re-optimized frequency curves where Case 1 tracks η_C from 0 to 0.857 and plateaus.